



NUST CHIP DESIGN CENTRE

Digital Logic Design

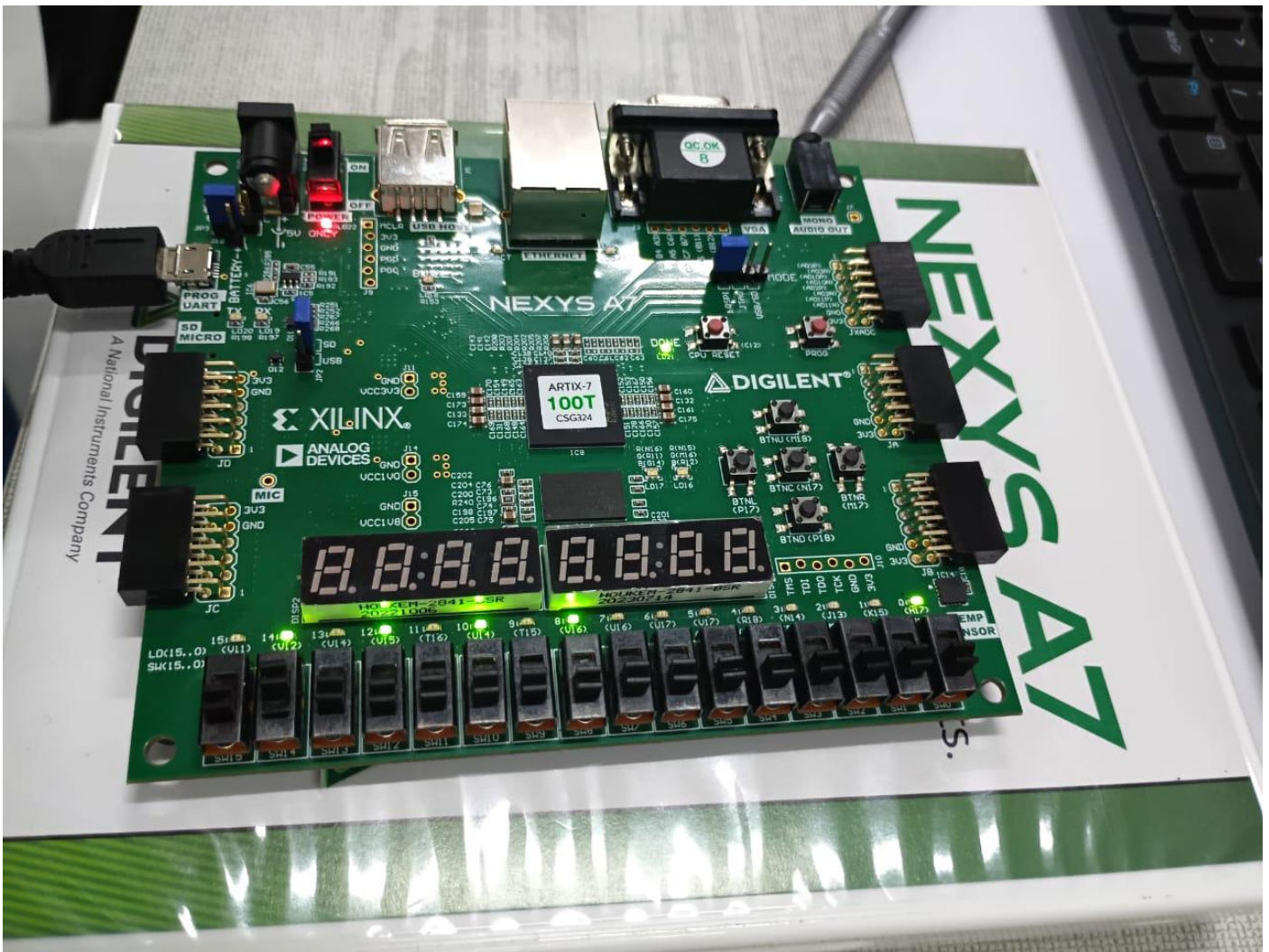
Lab Manual # 09

Sequential Circuits using System Verilog
(Counters / Digital clocks)

<u>Name</u>	<i><u>Muddassir Ali Siddiqui</u></i>
<u>Instructor</u>	<i><u>Sir Musaddiq Hussain & Sir Bilal</u></i>
<u>Date</u>	<i><u>2nd Aug 2025</u></i>

FPGA Outputs:

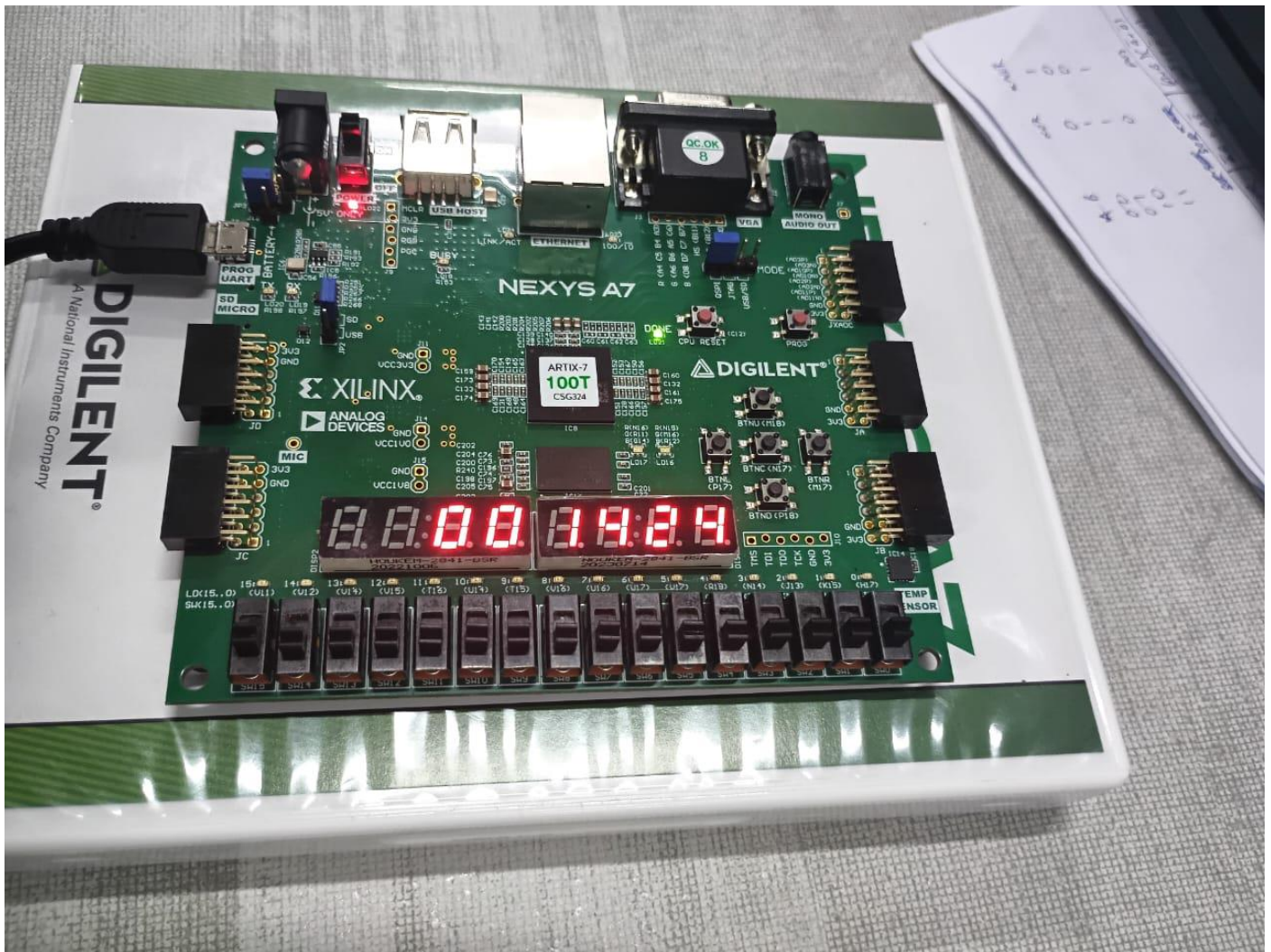
ii. Task 2:



iii. Task 3:

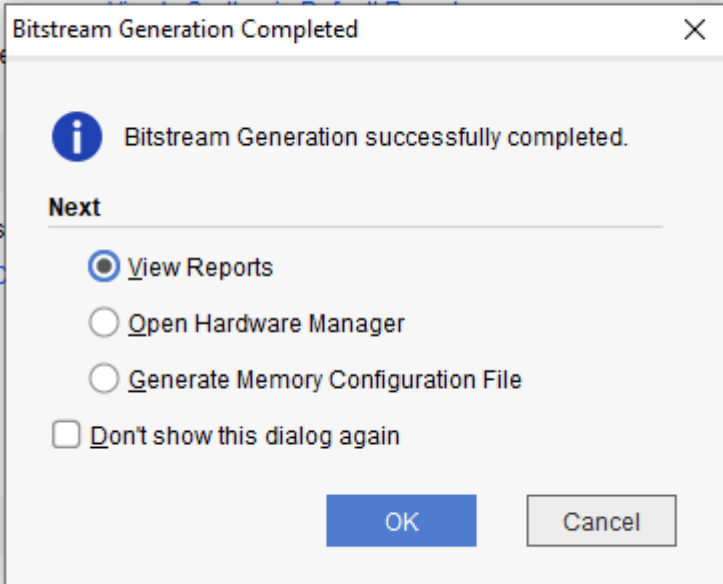


iv. Task 5:

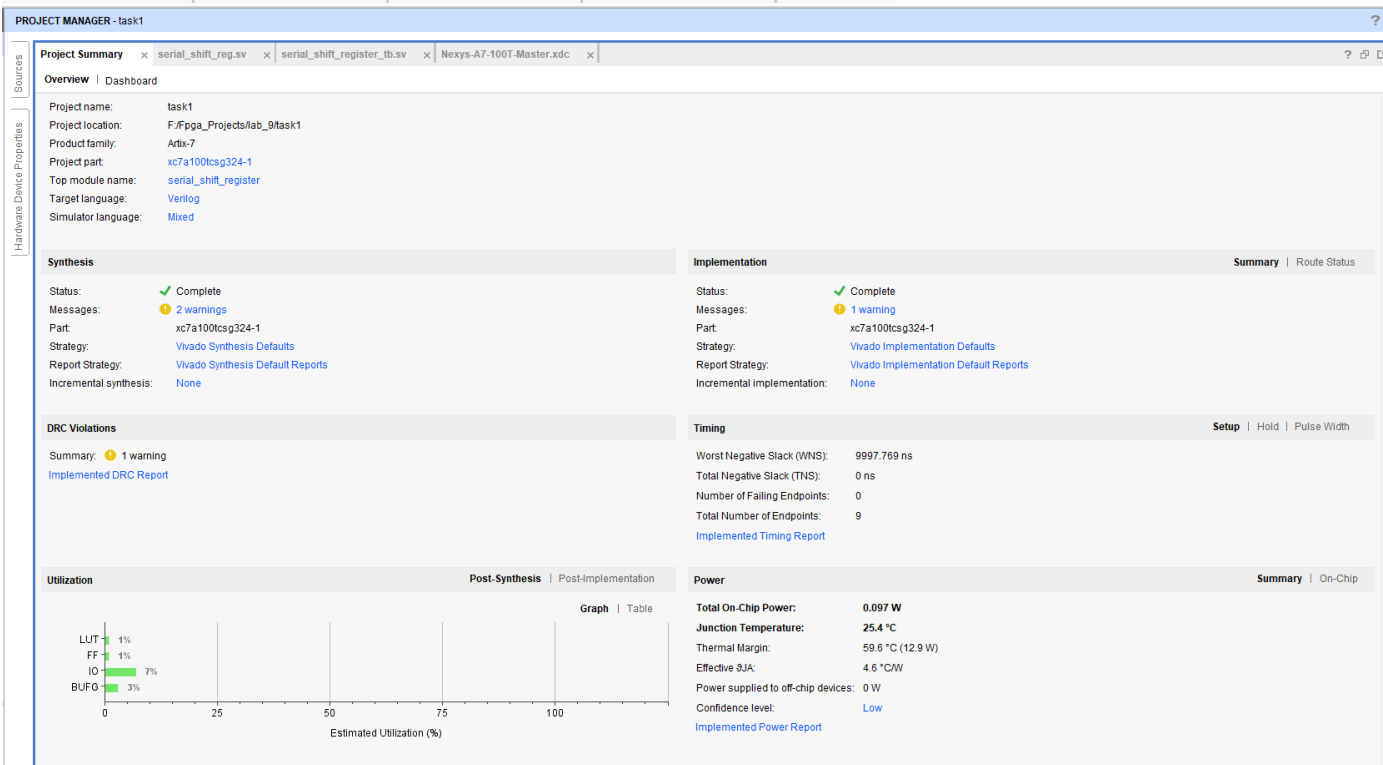


2. Vivado Screenshots:

Task 1:



The dialog box titled "Bitstream Generation Completed" features an information icon and the message "Bitstream Generation successfully completed." Below this, under the "Next" section, are four options: "View Reports" (selected with a radio button), "Open Hardware Manager", "Generate Memory Configuration File", and "Don't show this dialog again" (unchecked checkbox). "OK" and "Cancel" buttons are at the bottom.



The "PROJECT MANAGER - task1" window displays project details and implementation status. The "Overview | Dashboard" tab is active, showing project information and implementation results.

Project Summary	
Project name:	task1
Project location:	F:/Fpga_Projects/lab_9/task1
Product family:	Artix-7
Project part:	xc7a100tcsq324-1
Top module name:	serial_shift_register
Target language:	Verilog
Simulator language:	Mixed

Synthesis	
Status:	Complete
Messages:	2 warnings
Part:	xc7a100tcsq324-1
Strategy:	Vivado Synthesis Defaults
Report Strategy:	Vivado Synthesis Default Reports
Incremental synthesis:	None

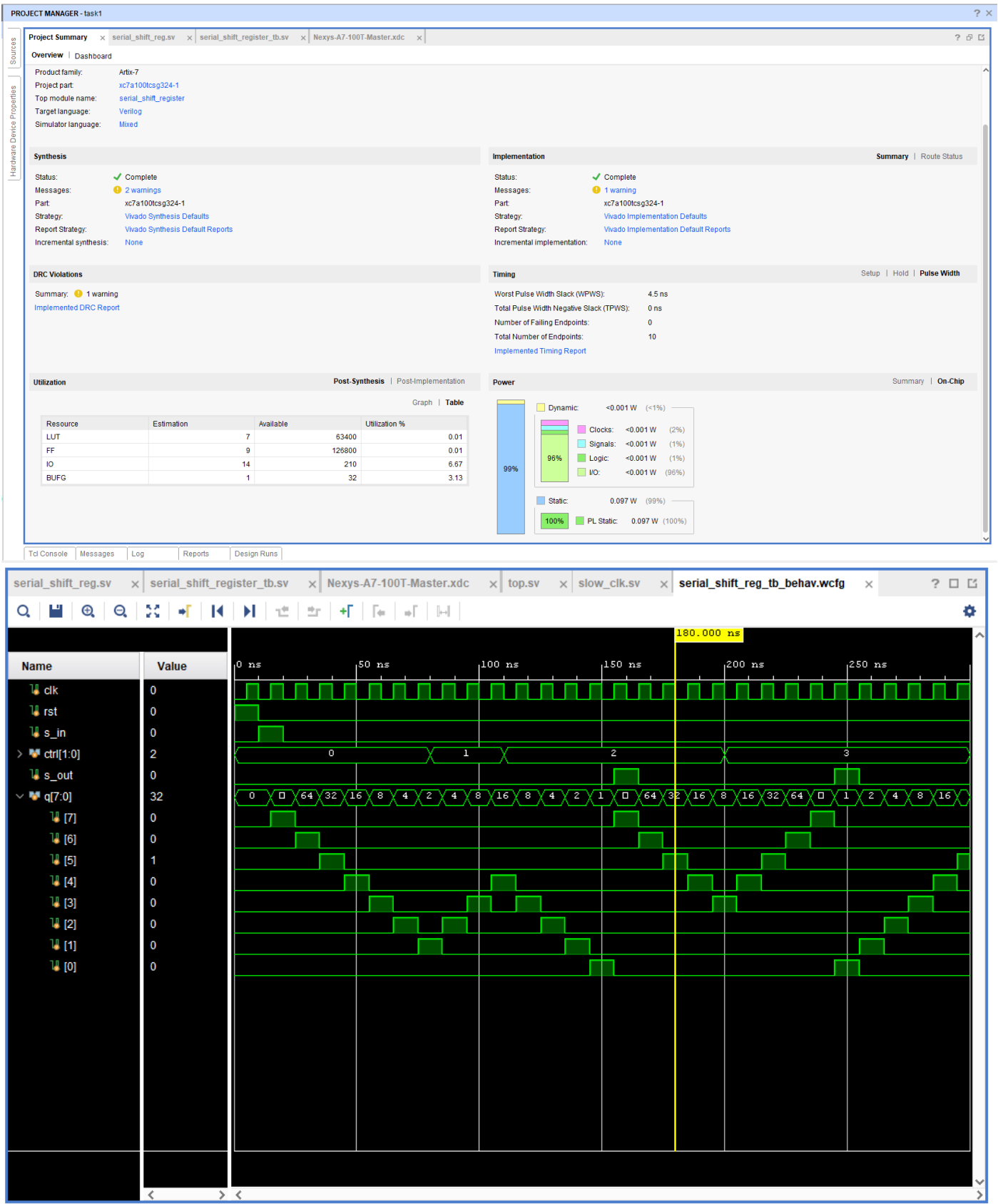
DRC Violations	
Summary:	1 warning
Implemented DRC Report	

Implementation	
Status:	Complete
Messages:	1 warning
Part:	xc7a100tcsq324-1
Strategy:	Vivado Implementation Defaults
Report Strategy:	Vivado Implementation Default Reports
Incremental implementation:	None

Timing	
Worst Negative Slack (WNS):	9997.769 ns
Total Negative Slack (TNS):	0 ns
Number of Failing Endpoints:	0
Total Number of Endpoints:	9
Implemented Timing Report	

Power	
Total On-Chip Power:	0.097 W
Junction Temperature:	25.4 °C
Thermal Margin:	59.6 °C (12.9 W)
Effective θ_{JA} :	4.6 °C/W
Power supplied to off-chip devices:	0 W
Confidence level:	Low
Implemented Power Report	

At the bottom, a "Utilization" bar chart shows: LUT 1%, FF 1%, IO 7%, and BUFG 3%. The x-axis is "Estimated Utilization (%)" from 0 to 100.



Task 2:

xc7a100tcsg324-1
Vivado Synthesis Defaults

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synthe

ns

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d DRC

Implementa

Graph | Ta

Bitstream Generation Completed

i Bitstream Generation successfully completed.

Next

☒ **V**iew Reports

☐ **O**pen Hardware Manager

☐ **G**enerate Memory Configuration File

☐ **D**on't show this dialog again

OK **Cancel**

PROJECT MANAGER - task5

Project Summary x top.sv x Universal_shift_reg.sv x top_tb.sv x Nexys-A7-100T-Master.xdc x

Overview | Dashboard

Project name: task5
Project location: F:\Ppga_Projects\lab_9\task5
Product family: Artix-7
Project part: xc7a100tcsg324-1
Top module name: top
Target language: Verilog
Simulator language: Mixed

Synthesis

Status: **Complete**
Messages: **1 warning**
Part: xc7a100tcsg324-1
Strategy: Vivado Synthesis Defaults
Report Strategy: Vivado Synthesis Default Reports
Incremental synthesis: None

Implementation

Status: **Complete**
Messages: No errors or warnings
Part: xc7a100tcsg324-1
Strategy: Vivado Implementation Defaults
Report Strategy: Vivado Implementation Default Reports
Incremental implementation: None

DRC Violations

No DRC violations were found.
[Implemented DRC Report](#)

Timing

Worst Negative Slack (WNS): 6.154 ns
Total Negative Slack (TNS): 0 ns
Number of Failing Endpoints: 0
Total Number of Endpoints: 27
[Implemented Timing Report](#)

Utilization

Post-Synthesis | **Post-Implementation**

Graph | Table

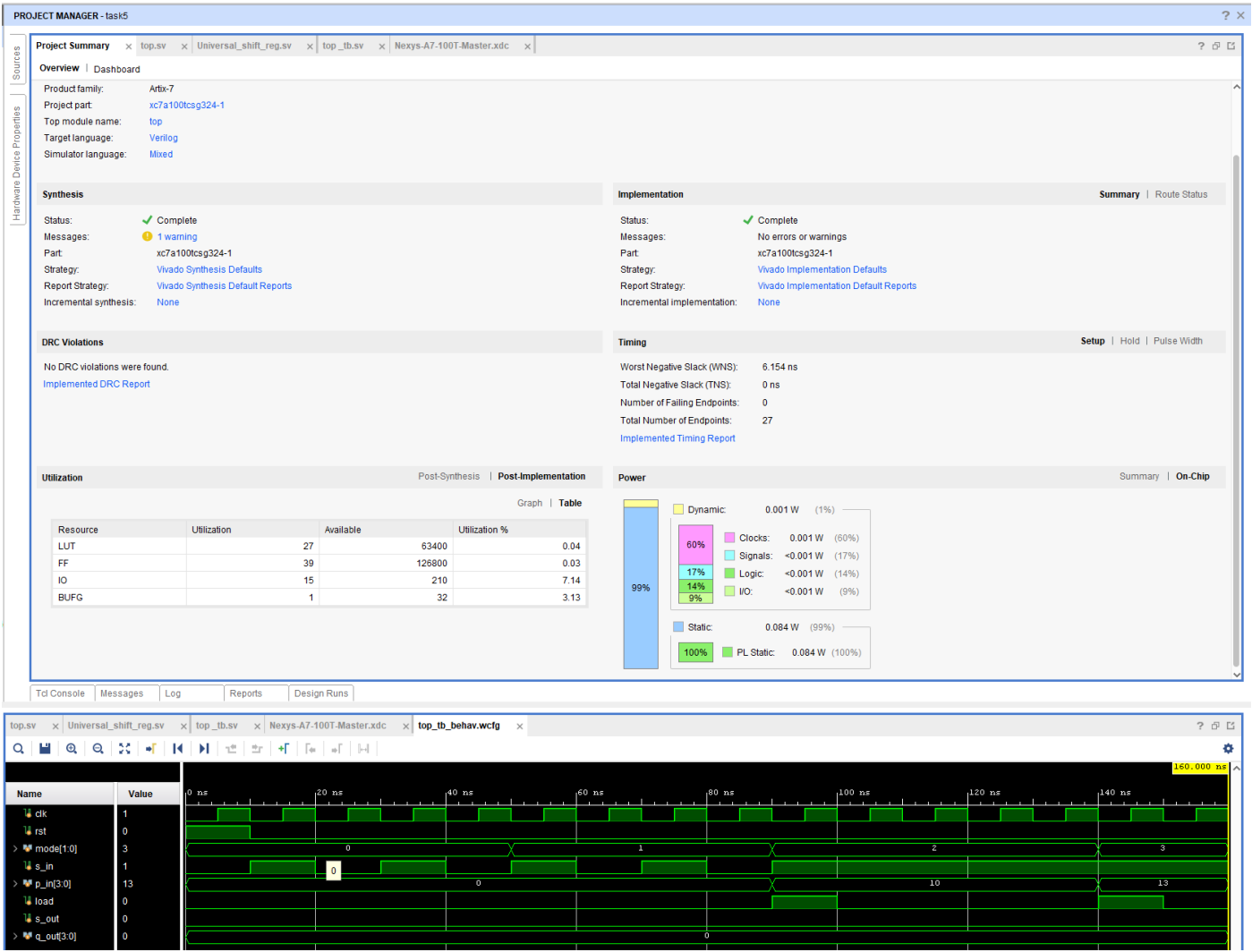
LUT: 1%
FF: 1%
IO: 7%
BUFG: 3%

Utilization (%)

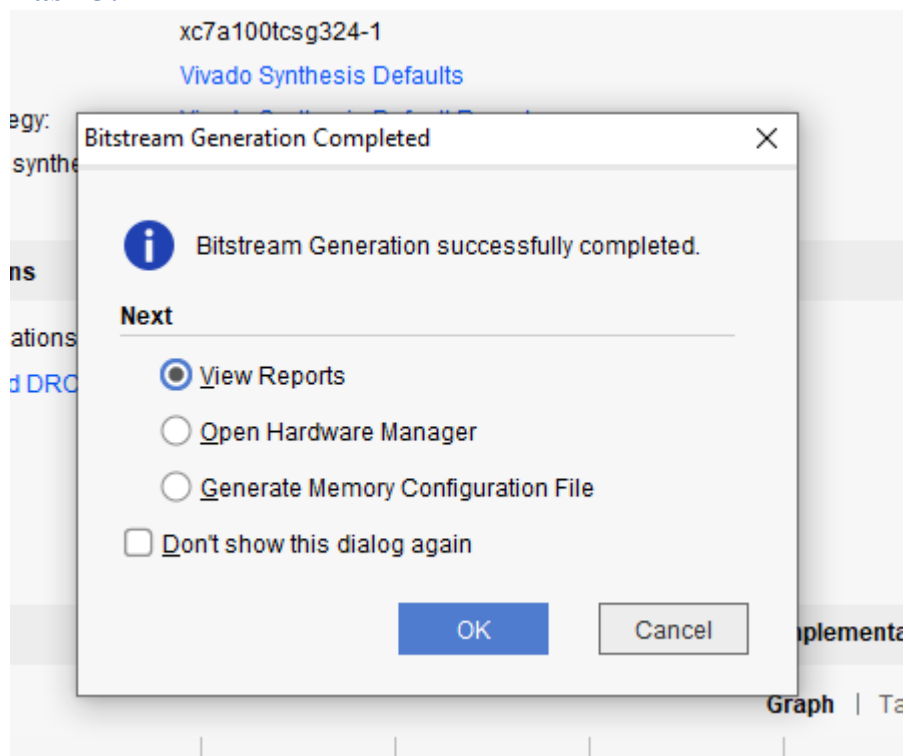
Power

Total On-Chip Power: 0.085 W
Junction Temperature: 25.4 °C
Thermal Margin: 59.6 °C (12.9 W)
Effective θ_{JA} : 4.6 °C/W
Power supplied to off-chip devices: 0 W
Confidence level: **Low**
[Implemented Power Report](#)

Tcl Console Messages Log Reports Design Runs



Task 3:



Project Summary

Overview | Dashboard

Product family: Artix-7
Project part: xc7a100tcsg324-1
Top module name: sequence_cntr
Target language: Verilog
Simulator language: Mixed

Synthesis

Status: ✔ Out-of-date
Messages: ⚠ 17 warnings
Part: xc7a100tcsg324-1
Strategy: Vivado Synthesis Defaults
Report Strategy: Vivado Synthesis Default Reports
Incremental synthesis: None

Implementation

Status: ✔ Complete
Messages: ⚠ 1 warning
Part: xc7a100tcsg324-1
Strategy: Vivado Implementation Defaults
Report Strategy: Vivado Implementation Default Reports
Incremental implementation: None

DRC Violations

Summary: ⚠ 1 warning
[Implemented DRC Report](#)

Timing

Worst Negative Slack (WNS): 5.952 ns
Total Negative Slack (TNS): 0 ns
Number of Failing Endpoints: 0
Total Number of Endpoints: 27
[Implemented Timing Report](#)

Utilization

Post-Synthesis | Post-Implementation

Graph | Table

LUT: 1%
FF: 1%
IO: 10%
BUFG: 3%

Utilization (%)

Power

Summary | On-Chip

Total On-Chip Power: 0.085 W
Junction Temperature: 25.4 °C
Thermal Margin: 59.6 °C (12.9 W)
Effective θJA: 4.6 °C/W
Power supplied to off-chip devices: 0 W
Confidence level: Medium
[Implemented Power Report](#)

Project Summary

Overview | Dashboard

Product family:Arrix-7

Project part:xc7a100tcs9324-1

Top module name:sequence_cntr

Target language:Verilog

Simulator language:Mixed

Synthesis

Status:Out-of-date

Messages:17 warnings

Part:xc7a100tcs9324-1

Strategy:Vivado Synthesis Defaults

Report Strategy:Vivado Synthesis Default Reports

Incremental synthesis:None

Implementation

Status:Complete

Messages:1 warning

Part:xc7a100tcs9324-1

Strategy:Vivado Implementation Defaults

Report Strategy:Vivado Implementation Default Reports

Incremental implementation:None

DRC Violations

Summary:1 warning

Implemented DRC Report

Timing

Worst Negative Slack (WNS):5.952 ns

Total Negative Slack (TNS):0 ns

Number of Failing Endpoints:0

Total Number of Endpoints:27

Implemented Timing Report

Utilization

Post-Synthesis | Post-Implementation

Graph | Table

Resource	Utilization	Available	Utilization %
LUT	26	63400	0.04
FF	36	126800	0.03
IO	20	210	9.52
BUFG	1	32	3.13

Power

Summary | On-Chip

Dynamic:0.001 W (1%)

Static:0.084 W (99%)

PL Static:0.084 W (100%)

sequence_cntr.sv | sequence_cntr_tb.sv | Nexys-A7-100T-Master.xdc | Untitled 19*

clk

rst

count[2:0]

0 ns

50 ns

100 ns

150 ns

190.000 ns

0

2

4

6

1

3

5

7

0

2

4

6

1

3

5

7

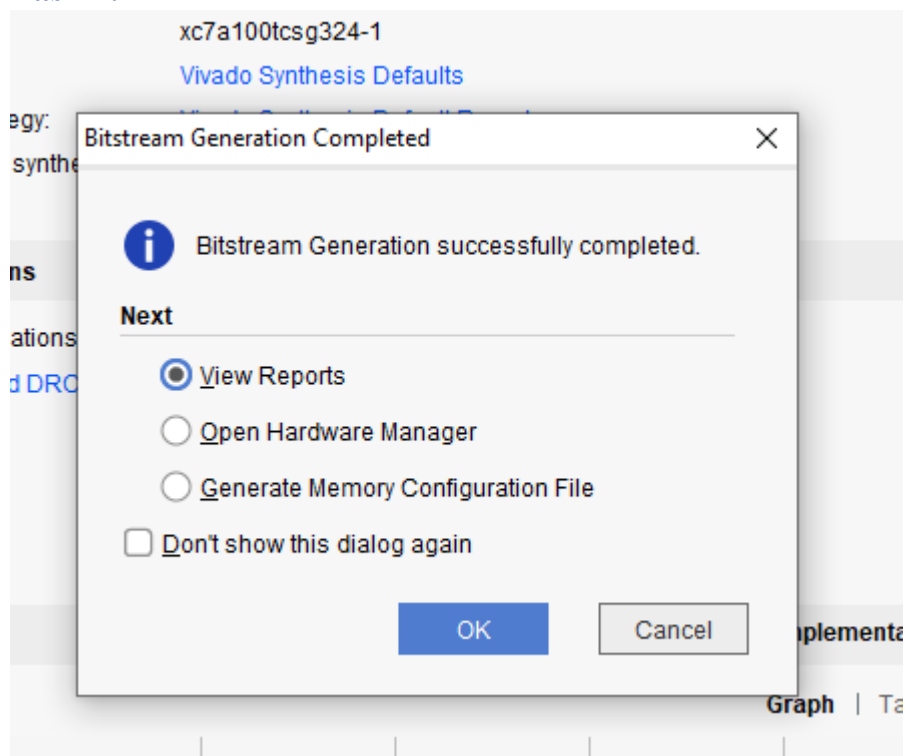
0

2

4

Nust Chip Design Center (NCDC)

Task 4:



Project Summary x sequence_cntr.v sv sequence_cntr_tb.v sv Nexys-A7-100T-Master.xdc x

Overview | Dashboard

Settings

Project name: task3
Project location: F:\Fpga_Projects\lab_9\task3
Product family: Artix-7
Project part: xc7a100tcsg324-1
Top module name: sequence_cntr
Target language: Verilog
Simulator language: Mixed

Synthesis

Status: Complete
Messages: 17 warnings
Part: xc7a100tcsg324-1
Strategy: Vivado Synthesis Defaults
Report Strategy: Vivado Synthesis Default Reports
Incremental synthesis: None

Implementation

Status: Complete
Messages: 1 warning
Part: xc7a100tcsg324-1
Strategy: Vivado Implementation Defaults
Report Strategy: Vivado Implementation Default Reports
Incremental implementation: None

DRC Violations

Summary: 1 warning
[Implemented DRC Report](#)

Timing

Worst Negative Slack (WNS): 5.952 ns
Total Negative Slack (TNS): 0 ns
Number of Failing Endpoints: 0
Total Number of Endpoints: 27
[Implemented Timing Report](#)

Utilization

Post-Synthesis | Post-Implementation

Graph | Table

Resource	Utilization (%)
LUT	1%
FF	1%
IO	10%
BUFG	3%

Power

Summary | On-Chip

Total On-Chip Power: 0.085 W
Junction Temperature: 25.4 °C
Thermal Margin: 59.6 °C (12.9 W)
Effective θJA: 4.6 °C/W
Power supplied to off-chip devices: 0 W
Confidence level: Medium
[Implemented Power Report](#)

Tcl Console Messages Log Reports Design Runs

Project Summarysequence_cntr.svsequence_cntr_tb.svNexys-A7-100T-Master.xdc

OverviewDashboard

Product family:Artix-7
Project part:xc7a100tcsg324-1
Top module name:sequence_cntr
Target language:Verilog
Simulator language:Mixed

Synthesis

Status:Complete
Messages:17 warnings
Part:xc7a100tcsg324-1
Strategy:Vivado Synthesis Defaults
Report Strategy:Vivado Synthesis Default Reports
Incremental synthesis:None

DRC Violations

Summary:1 warning
Implemented DRC Report

Implementation

Status:Complete
Messages:1 warning
Part:xc7a100tcsg324-1
Strategy:Vivado Implementation Defaults
Report Strategy:Vivado Implementation Default Reports
Incremental implementation:None

Timing

Worst Negative Slack (WNS):5.952 ns
Total Negative Slack (TNS):0 ns
Number of Failing Endpoints:0
Total Number of Endpoints:27
Implemented Timing Report

Utilization

Post-SynthesisPost-Implementation
GraphTable

Resource	Utilization	Available	Utilization %
LUT	26	63400	0.04
FF	36	126800	0.03
IO	20	210	9.52
BUFG	1	32	3.13

Power

SummaryOn-Chip
Dynamic:0.001 W (1%)
64% Clocks:0.001 W (64%)
16% Signals:<0.001 W (16%)
14% Logic:<0.001 W (14%)
14% IO:<0.001 W (6%)
Static:0.084 W (99%)
100% PL Static:0.084 W (100%)

clk_div_25hz.svclk_div_25hz_tb.svUntitled 31*

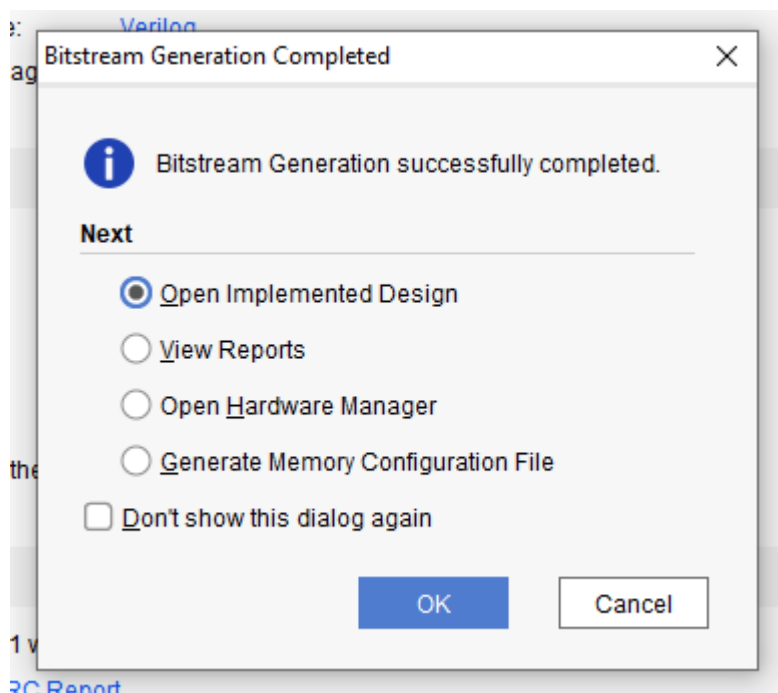
clk_100MHz1
rst0
clk_25MHz1
clk_75kHz1

0 us5 us10 us15 us20 us25 us30 us35 us40 us

0

Nust Chip Design Center (NCDC)

Task 5:



RC Report

Project Summary | digital_clk.sv | top.sv | Nexys-A7-100T-Master.xdc | digital_clk_tb.sv

Overview | Dashboard

Project name: task5
Project location: F:\Fpga_Projects\lab_9\task5
Product family: Artix-7
Project part: xc7a100tcsq324-1
Top module name: top
Target language: Verilog
Simulator language: Mixed

Synthesis

Status: ✓ Complete
Messages: 3 warnings
Part: xc7a100tcsq324-1
Strategy: Vivado Synthesis Defaults
Report Strategy: Vivado Synthesis Default Reports
Incremental synthesis: None

Implementation

Status: ✓ Complete
Messages: No errors or warnings
Part: xc7a100tcsq324-1
Strategy: Vivado Implementation Defaults
Report Strategy: Vivado Implementation Default Reports
Incremental implementation: None

DRC Violations

No DRC violations were found.
[Implemented DRC Report](#)

Timing

Worst Negative Slack (WNS): 5.855 ns
Total Negative Slack (TNS): 0 ns
Number of Failing Endpoints: 0
Total Number of Endpoints: 48
[Implemented Timing Report](#)

Utilization

Post-Synthesis | Post-Implementation

Graph | Table

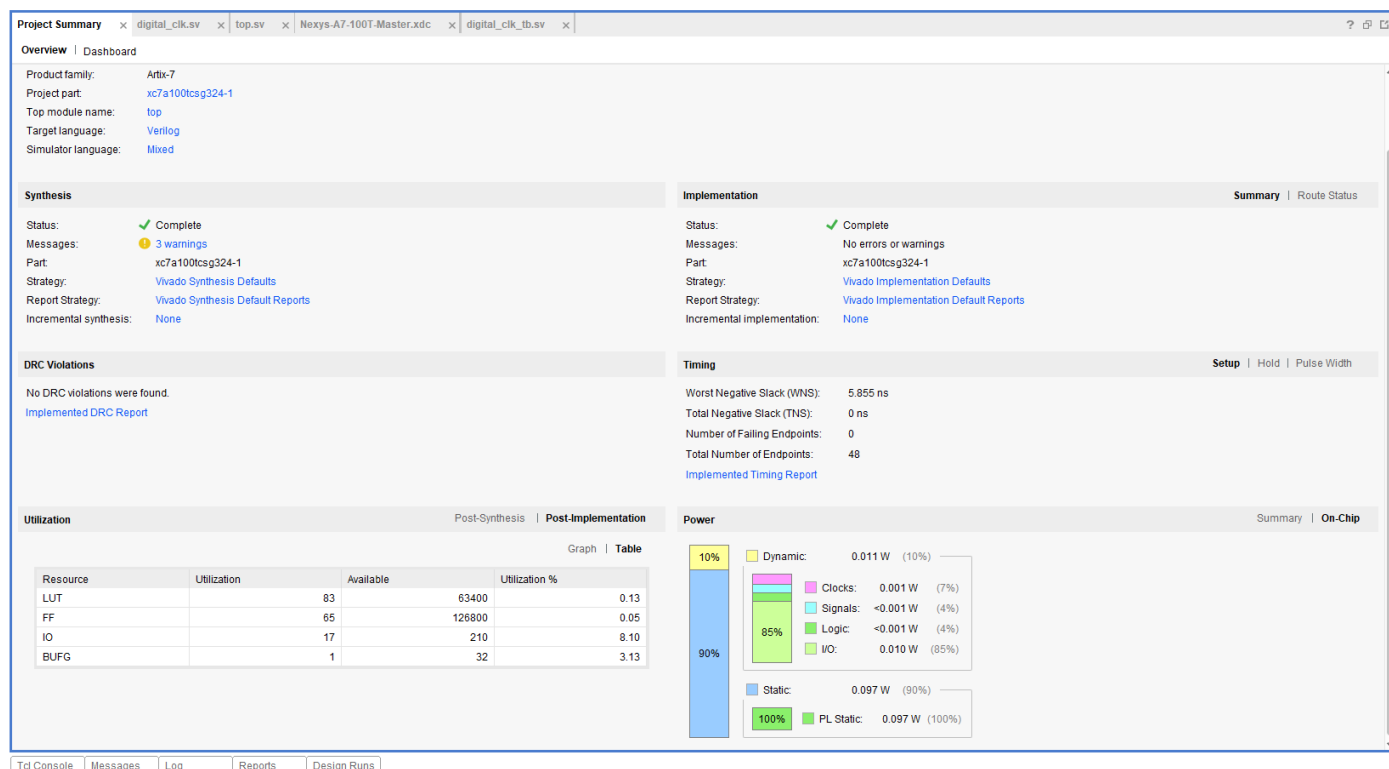
Resource	Utilization (%)
LUT	4%
FF	4%
IO	8%
BUFG	3%

Power

Summary | On-Chip

Total On-Chip Power: 0.108 W
Junction Temperature: 25.5 °C
Thermal Margin: 59.5 °C (12.9 W)
Effective SJA: 4.6 °C/W
Power supplied to off-chip devices: 0 W
Confidence level: Low
[Implemented Power Report](#)

Tcl Console | Messages | Log | Reports | Design Runs



3. Critical Analysis: (Write you critical analysis / conclusion here)

In this lab we design in task 1 design a serial in and serial out and in task 2 design a universal shift registers with mux with 4 different shift operation pipo, pipo, siso, sipo. In task 3 design a counter and in task 4 we design clk freq divider circuit. By the way I use this freq divider circuit in every lab task so that I could see the operations with the naked eye. And in the last we design a digital clock and display it on seven segments.